

Matthew Minwoo Kim

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Education

Seoul National University <i>B.S. in Computer Science and Engineering</i> Current GPA: 4.21/4.3 (67 credits)	<i>Seoul, South Korea</i> <i>2023 – Present</i>
Korea Science Academy of KAIST <i>High School Diploma</i> GPA: 4.14/4.3 (top 10% of the class)	<i>Busan, South Korea</i> <i>2020 – 2023</i>
University of Cambridge <i>International Summer School</i> Business and Entrepreneurship course	<i>Cambridge, UK</i> <i>Summer 2022</i>

Research Experience

Undergraduate Research Opportunity Program (UROP) <i>IDS Lab, Seoul National University</i> - Supervised under Prof. Sang-goo Lee - Research Topic: "Text-to-SQL Generation using Intermediate Representation"	<i>Seoul, South Korea</i> <i>Jul 2024 – Present</i>
Cryptography Research <i>Korea Science Academy of KAIST</i> - Research Topic: "A Secret Sharing Scheme without Third-Party Dealers" - Resulted in Regeneron ISEF (International Science and Engineering Fair) South Korea Finalist	<i>Busan, South Korea</i> <i>2021 – 2022</i>

Professional Experience

AI System Developer <i>Republic of Korea Air Force</i> Researching and developing military-specific Computer Vision and Speech-to-Text (STT) systems	<i>South Korea</i> <i>Oct 2024 – Present</i>
Research Team Intern <i>AIBIZON AI LAB</i> - Developed a CRDT-based program for synchronous connection between Slack and a database - Keywords: TypeScript, CRDT, Prisma, Database - Conducted an empirical study on RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval)	<i>Online</i> <i>Mar 2024 – Apr 2024</i>
Research & Infra Team Intern <i>Hyperithm</i> - Developed & maintained trading strategy tool - Keywords: Asynchronous Programming, TypeScript, Grafana, Python - Developed HFT-optimized cryptocurrency exchange data feed - Keywords: Rust, Asynchronous Programming	<i>South Korea</i> <i>Jun 2023 – Aug 2023</i>

Awards & Achievements

Regeneron ISEF (International Science and Engineering Fair) Systems Software category, South Korea Finalist (Team Leader)	<i>2022</i>
R&E Project Presentation Contest for the Scientific Talent Grand Prize (Ministry of Science and ICT Award)	<i>2021</i>

Certificates

Craftsman Information Processing (Apr 2024)

SQL Developer (Jun 2024)

Research Interests

- Natural Language Processing
- Computer Vision
- Multimodal
- Cryptography